

$$\underbrace{\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}}_{\hat{y}: n \times 1} = \underbrace{\begin{bmatrix} \textcolor{brown}{1} & x_{11} & x_{12} & \cdots & x_{1m} \\ \textcolor{brown}{1} & x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \vdots & & \vdots \\ \textcolor{brown}{1} & x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}}_{X: n \times (m+1)} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix}}_{\beta: (m+1) \times 1}$$

an extra column of 1s carries the intercept β_0

rows: data points (n) columns: a 1, then the m inputs β : the $m + 1$ numbers to find